



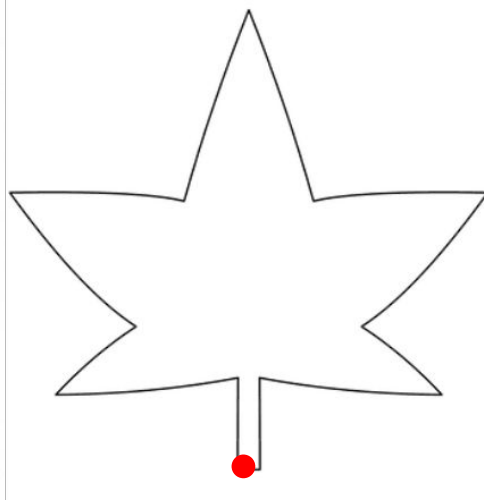
Look! A point



Point



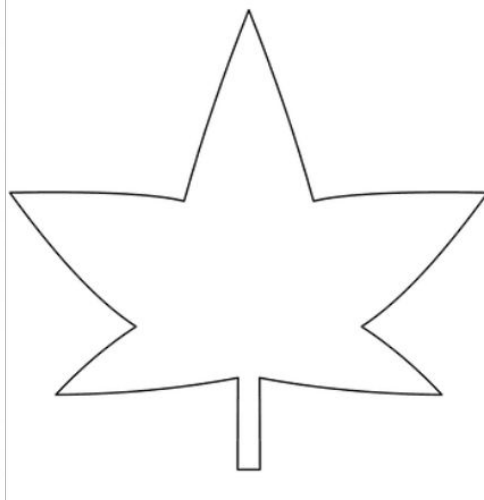
Woah! It drew a shape.



Shape



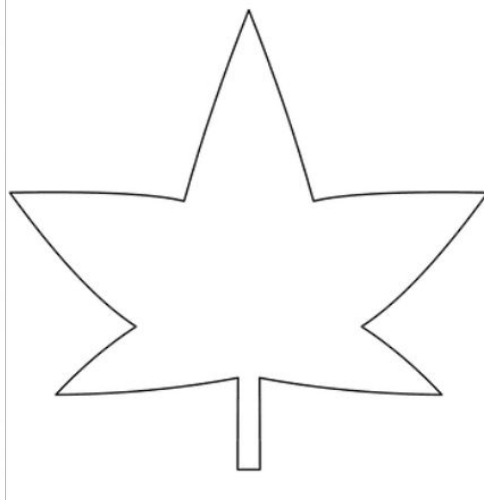
Let's look at this shape closely.



Shape



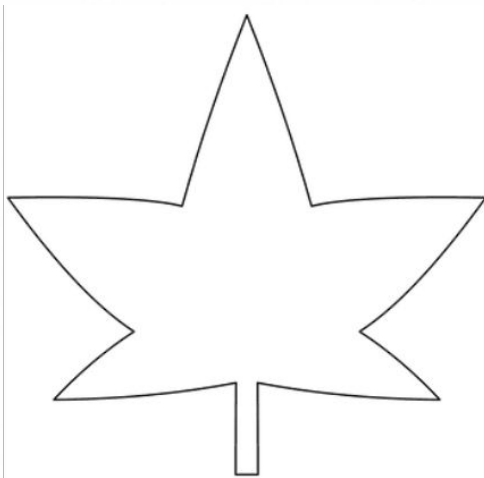
There are no gaps in its boundary.



Shape



Is it open or closed?

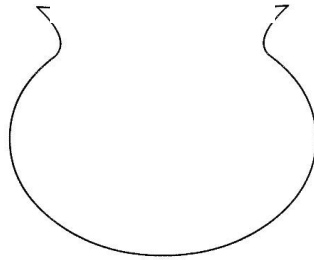


Open

Closed



Is this shape open or closed?



Open

Closed



What about this one?



Open

Closed



And this?

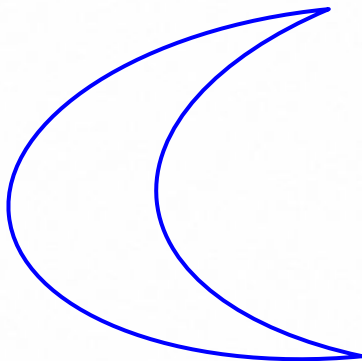


Open

Closed



How about this one?



Open

Closed



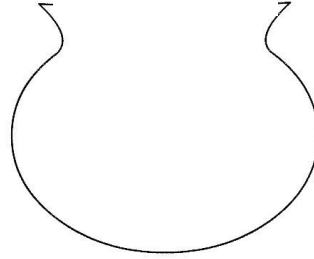
These two shapes are open and these two shapes are closed.



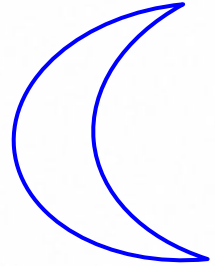
Closed



Open



Open



Closed



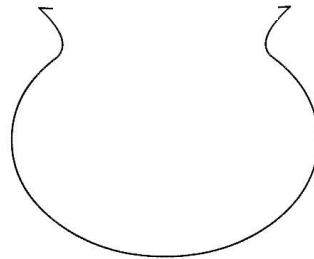
Look! The boundaries of the shapes are different too.



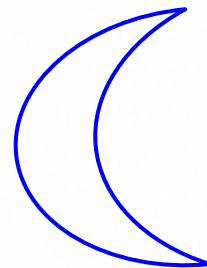
Closed



Open



Open



Closed



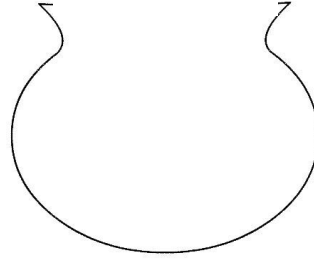
Some are straight and some are curved.



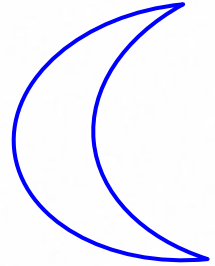
Closed



Open



Open



Closed



Which boundaries are straight and which are curved?



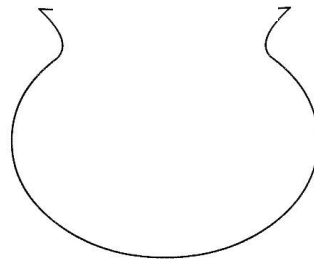
Closed

Straight ▼



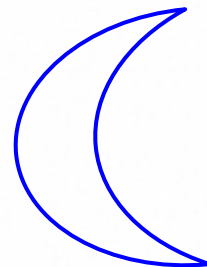
Open

Straight ▼



Open

Curved ▼



Closed

Curved ▼



You got it! These boundaries are straight, while these are curved.



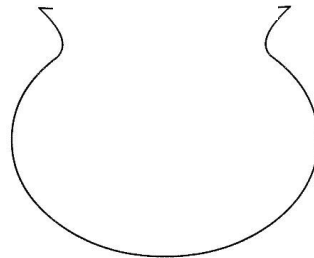
Closed

Straight



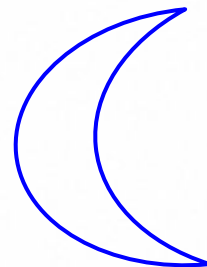
Open

Straight



Open

Curved

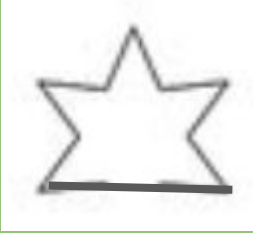


Closed

Curved



Tap the figure which is both closed and made of straight lines.



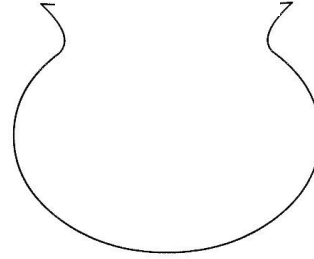
Closed

Straight



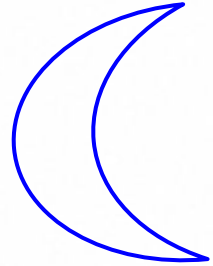
Open

Straight



Open

Curved



Closed

Curved



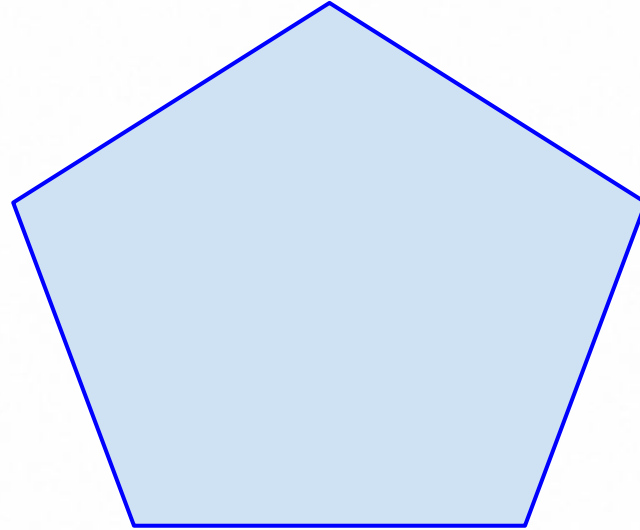
We call this a polygon.



Polygon

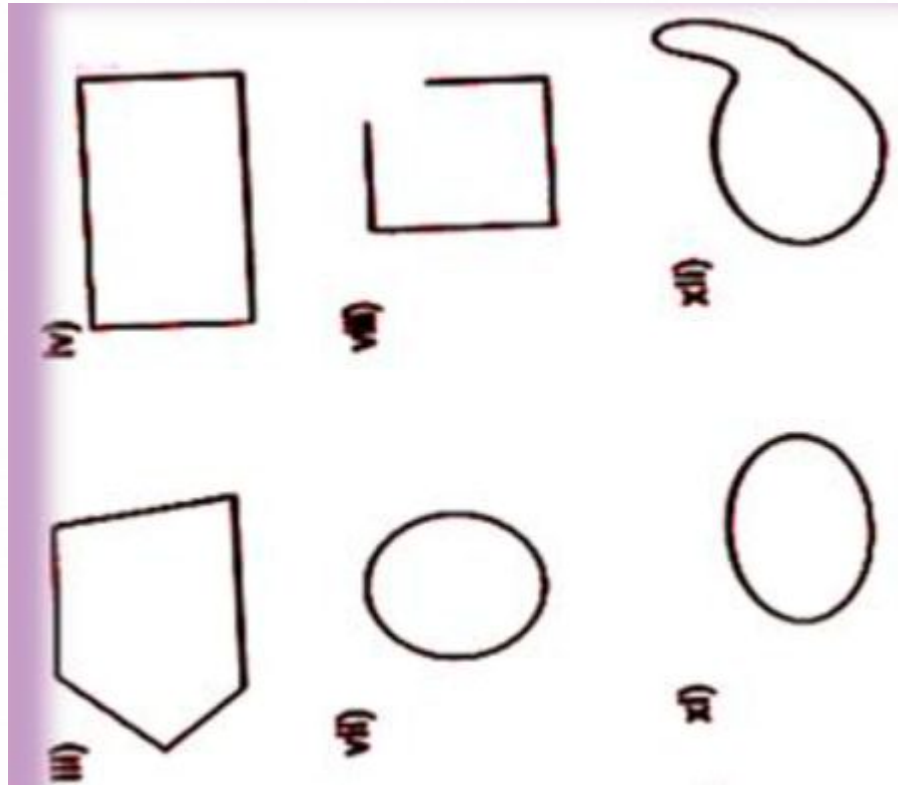


A closed figure made
of only straight line
segments is called a
polygon



Polygon

Which of these figures are polygons?



Check

1.9

Check Your Understanding

Select all the figures that are polygons.

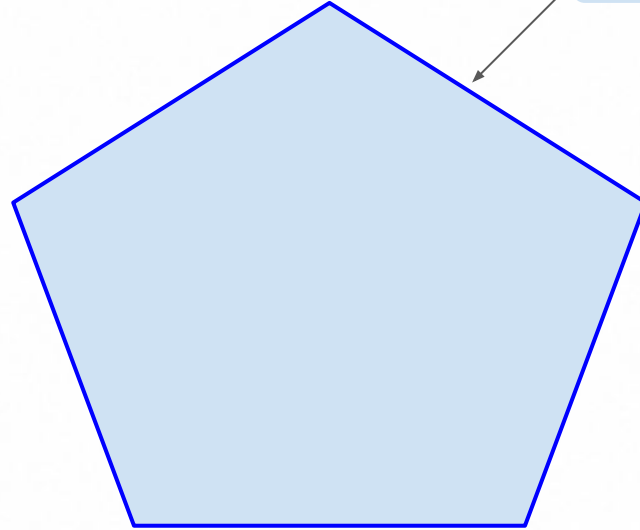
1 ☐2 ☐3 ☐4 ☐5 ☐

Remember: A polygon is closed and made only of straight line segments.



These line segments
are the sides of this
polygon.

Sides



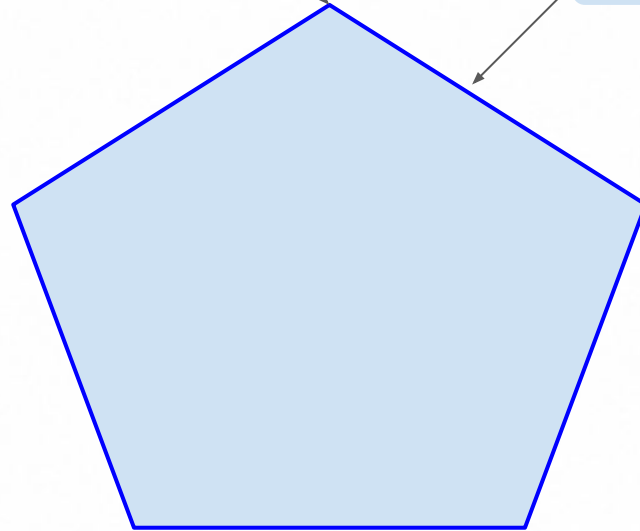
Polygon

The point where two sides meet is called a vertex.

Vertex

Sides

Polygon



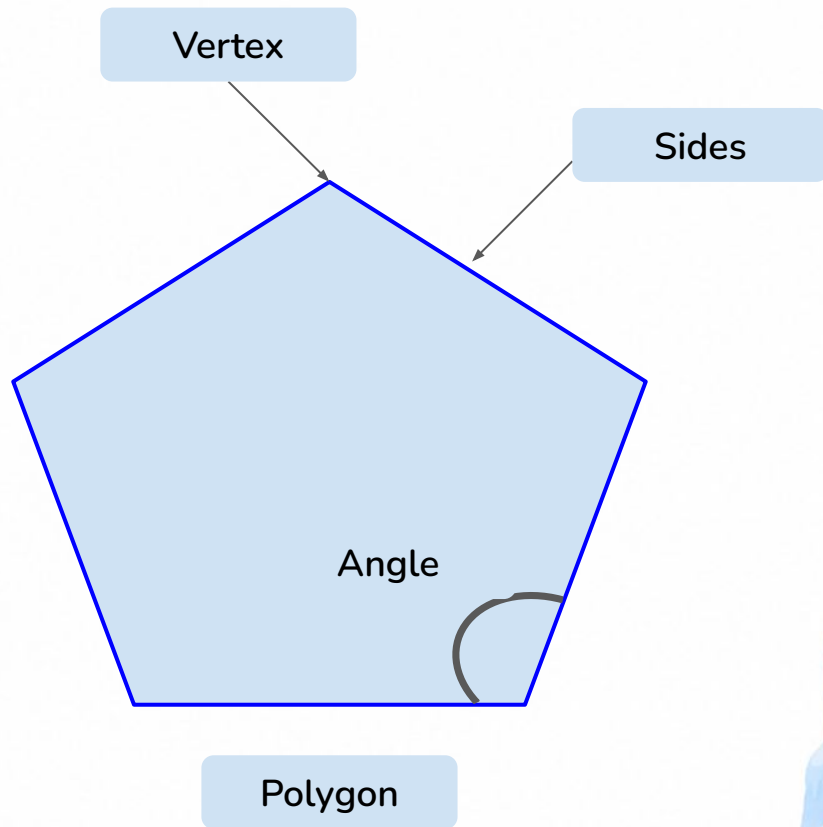
When two sides meet
they also form an
angle.

Vertex

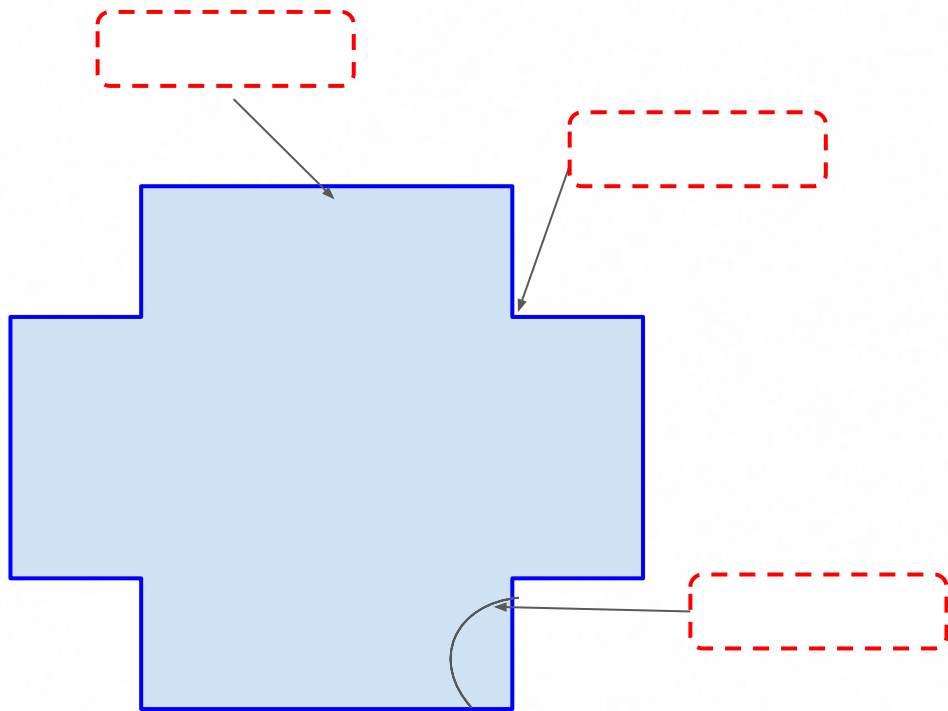
Sides

Angle

Polygon



Label the parts of the polygon.



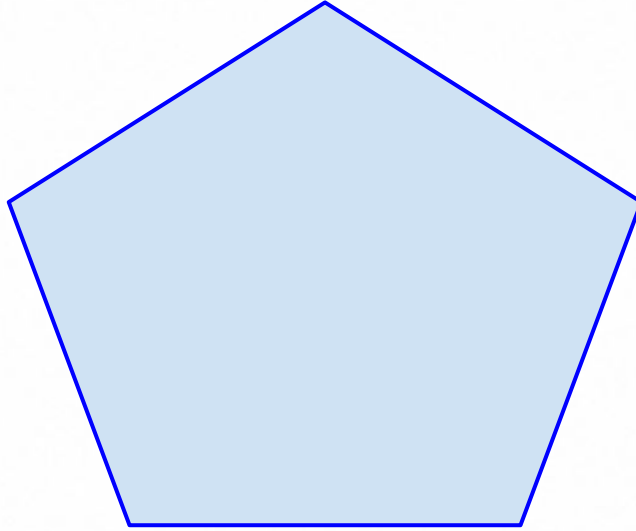
Side

Vertex

Angle



How many sides does this polygon have?



Polygon

Number of
sides

=

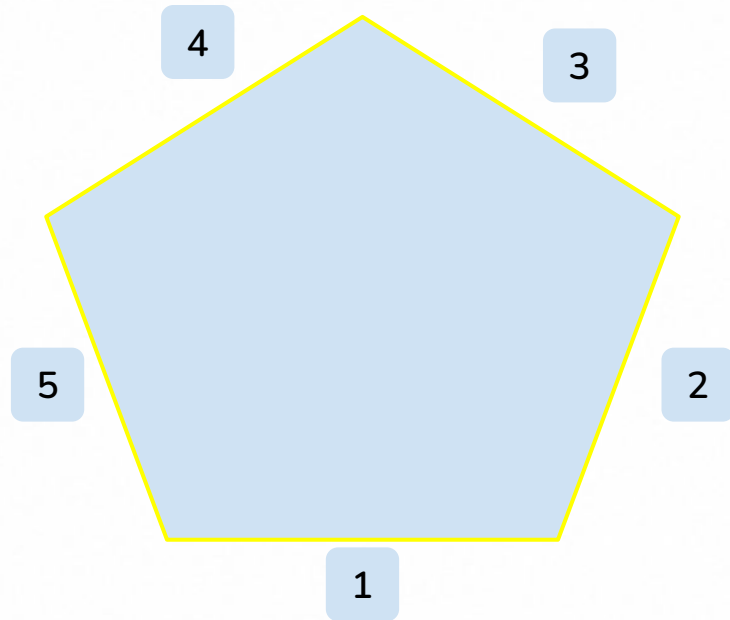
4



Check



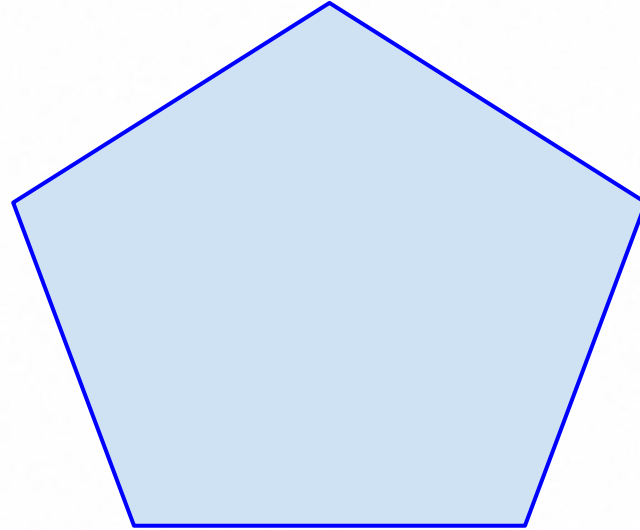
This polygon has 5 sides.



Polygon



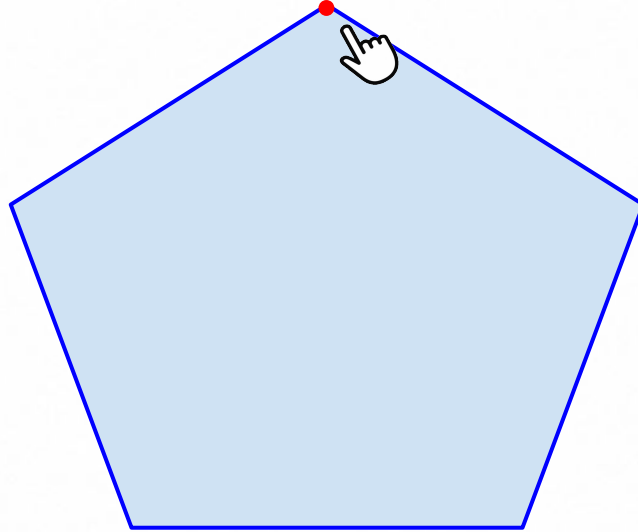
Now let's change how it looks.



Polygon



Drag any vertex. Stretch it, squash it or resize it.



Polygon





Whoa! It looks quite different now.

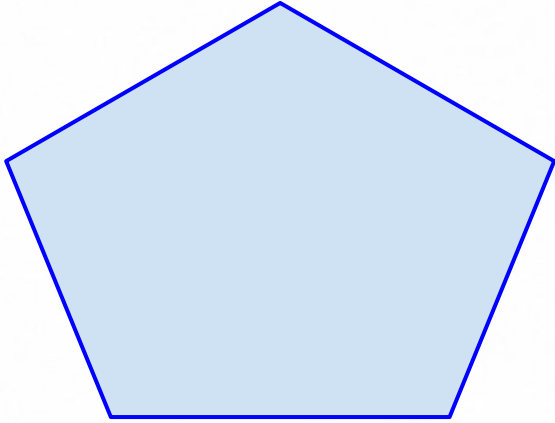


Polygon





It's appearance has completely changed.



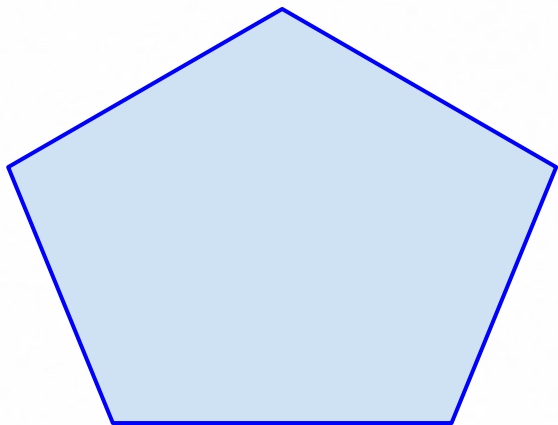
Before



After



But not everything has changed.



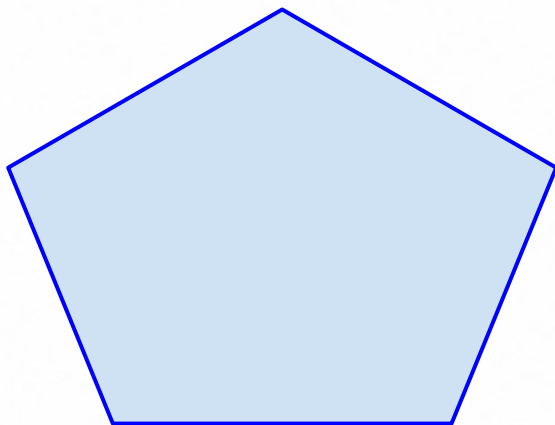
Before



After



Let's count the sides again.



Before

Number of
sides

5



After

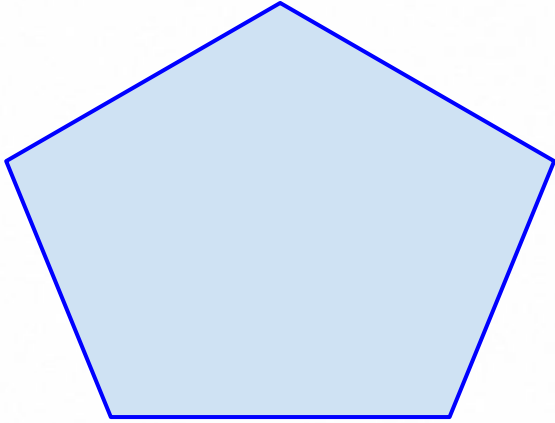
5



Check



Both polygons still have 5 sides.



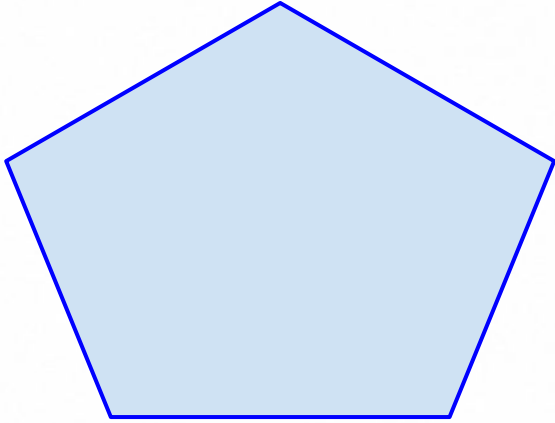
Before



After



So the number of sides gives us a good way to name polygons.



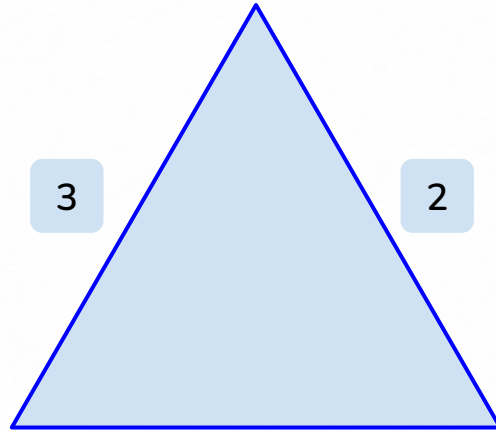
Before



After



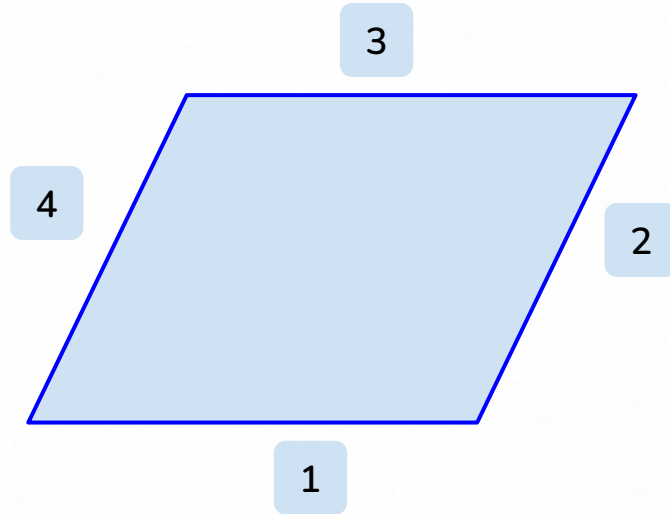
A polygon with 3 sides is called a triangle.



Triangle

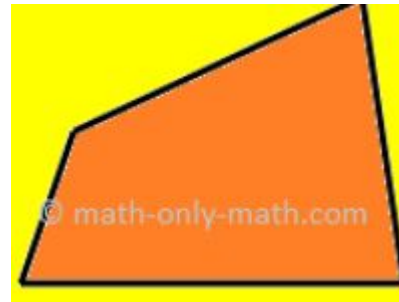
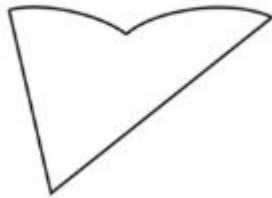
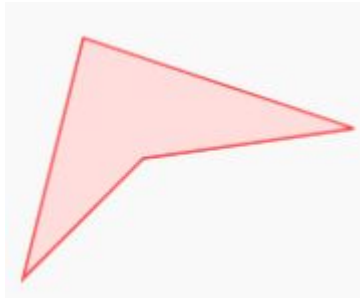


A polygon with 4 sides is called a quadrilateral.



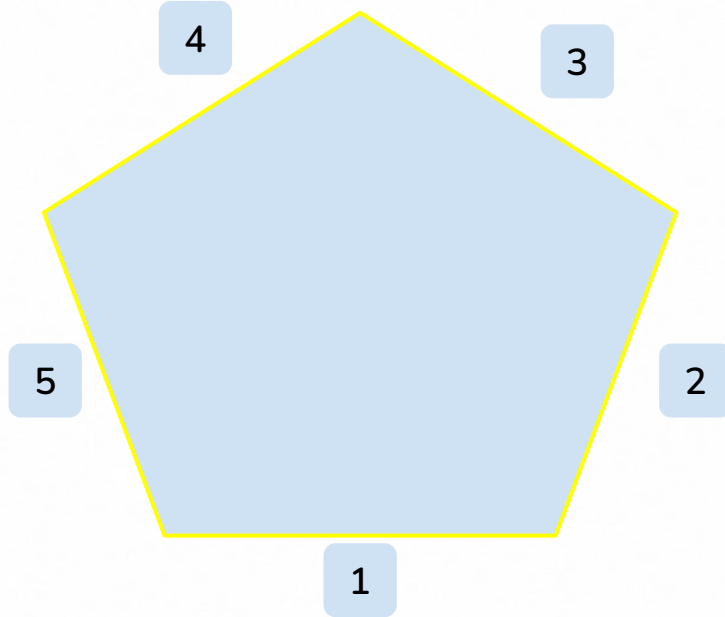
Quadrilateral

Select all the quadrilaterals.





A polygon with 5 sides is called a Pentagon.



Pentagon



1 2 3



A polygon with 3 sides
is called a **TRIANGLE**.



1 2 3 4



A polygon with 4 sides
is called a **QUADRILATERAL**.



1 2 3 4 5



A polygon with 5 sides
is called a **PENTAGON**.



1 2 3 4 5 6

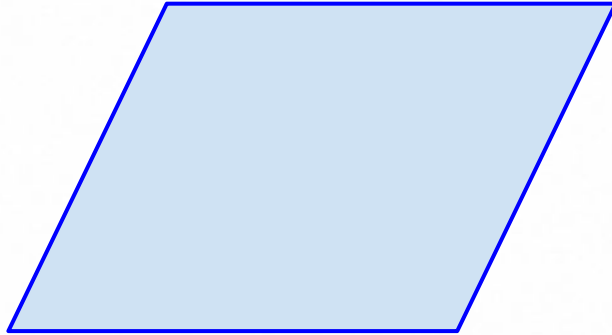


A polygon with 6 sides
is called a **HEXAGON**.

...and so on for more polygons.



Let's recall the different types of polygons we learnt about.



Quadrilateral

Sides

-

4

+



CHECK YOUR UNDERSTANDING

1 Which of these is a polygon?

Select all that are polygons.



A



B



C



D



E

3 Find the odd one out.

Which figure is NOT a polygon?



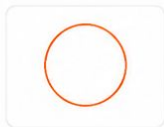
A



B



C



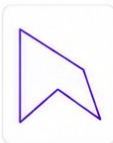
D

4 Which of these is a pentagon?

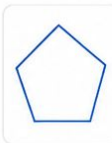
Select all that are pentagons. (Includes convex and concave examples)



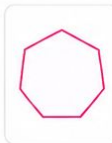
A



B



C



D



E



F

2 Classify the figures.

Drag each figure to the correct column.

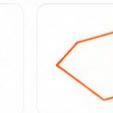


POLYGONS

NOT POLYGONS

5 Classify each figure as a Hexagon or a Heptagon.

Drag each figure to the correct column.



HEXAGON (6 sides)

HEPTAGON (7 sides)

CFU	Correct feedback	Incorrect feedback – Attempt 1	Incorrect feedback – Attempt 2
1. Which of these is a polygon?	That is a polygon! It is closed and made only of straight line segments.	Not quite. Check if the figure is closed and made only of straight line segments.	Remember, a polygon must be closed and have only straight sides. Try again.
2. Classify as Polygon / Not a Polygon	Great job! You sorted the figures correctly.	Some figures are in the wrong group. Check for gaps or curved sides.	Look closely. Polygons are closed and made only of straight line segments. Try sorting them again.
3. Find the odd one out	You found it! This figure is not a polygon.	Look again. Which figure does not follow the rule for a polygon?	Check each figure. Is it closed? Is it made only of straight line segments?
4. Which of these is a pentagon?	Exactly! A pentagon has 5 sides, even when its shape looks different.	Not quite. Count the sides of each figure carefully.	A pentagon always has 5 sides. Look for every figure with exactly 5 sides.
5. Classify as Hexagon / Heptagon	Well done! Hexagons have 6 sides and heptagons have 7 sides.	Almost! Count the sides of the figures you placed incorrectly.	Remember: 6 sides make a hexagon and 7 sides make a heptagon. Try sorting them again.